Midterm

Documentation

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Java Programming (COMP228)

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19 October 2026

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Class Hierarchy:

I created single, multilevel, and hierarchical inheritance to organize patients and staff. Java will not extend multiple classes, so I combined regular class inheritance with interfaces to build a hybrid structure.

Interfaces:

Instead of inheriting from multiple classes, I made and used the Reportable and Schedulable interfaces. They act as rules/contracts that make sure classes (like SpecialistDoctor) can handle reports and schedule appointments.

Static Variables:

These are shared counters, like personCount and patientCount, that belong to the class itself instead of any single object. They make it easy to track total registrations across the entire project.

Final Variables:

Final variables like ID numbers and fixed fees are unchangeable constants. Once they are set through the constructor, Java will not let them be changed, which keeps important data safe from accidental edits.

Method Overloading:

This allows use of the same method name, like calculating consultation fees or registering patients, multiple times as long as a different parameter list is given. It keeps the code flexible and easy to read.

Method Overriding:

Subclasses can rewrite methods inherited from their parents so they behave correctly for that specific role. This is what drives runtime polymorphism, and lets Java pick the right method to run when superclass is used.

Abstract Methods:

Abstract things are like templates that cannot be created directly with new. I made certain methods abstract to make every subclass fill in their own required details.